

1. Windows.System.Pslist

Description

The **Pslist** artifact enumerates all active processes executing on the system and provides fundamental process metadata such as Process ID (PID), Parent Process ID (PPID), executable path, creation time, command line, user context, and process hierarchy. It serves as the foundation for process-centric memory analysis.

Why Selected for Memory Forensics

Process information forms the basis of almost every forensic investigation. Nearly all memory artifacts, including handles, threads, DLLs, and VAD regions, are associated with a process. Pslist enables correlation between these artifacts by providing process identities and execution context.

Unique Contribution

Although Sysmon records process creation events, it does not provide a complete snapshot of the current process state. Pslist captures the processes that are actively resident in memory at collection time, enabling investigation of surviving malicious processes and process relationships.

Forensic Intelligence Provided

- Active malicious processes
 - Parent-child process hierarchy
 - Process execution context
 - Process lifetime
 - Suspicious executable locations
 - Foundation for cross-artifact correlation
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2. Windows.Network.Netstat

Description

The Netstat artifact enumerates active TCP and UDP network connections, listening ports, remote endpoints, and their associated processes.

Why Selected for Memory Forensics

Although network logs capture communication events, Netstat provides the live network state associated with running processes. It links active memory-resident processes with ongoing network activity.

Unique Contribution

Network telemetry records communication history, whereas Netstat identifies which memory-resident process currently owns an active connection. This enables direct correlation between process memory evidence and live network activity.

Forensic Intelligence Provided

- Active C2 connections
 - Suspicious remote endpoints
 - Process-to-network mapping
 - Persistence through network communication
 - Lateral movement indicators
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3. Windows.System.Drivers

Description

This artifact enumerates all kernel drivers currently loaded into memory together with their paths, signatures, and metadata.

Why Selected for Memory Forensics

Kernel drivers represent one of the highest privilege levels within Windows. Many advanced threats deploy malicious or vulnerable drivers to bypass security controls or obtain kernel-level execution.

Unique Contribution

Sysmon records driver loading events but cannot determine which drivers remain resident after execution. Memory analysis identifies drivers currently loaded within kernel memory.

Forensic Intelligence Provided

- Unsigned drivers
- Suspicious driver locations
- Rootkit indicators

- Kernel persistence
 - Driver-based privilege escalation
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4. Windows.System.DLLs

Description

The DLL artifact enumerates all dynamic-link libraries loaded by every running process.

Why Selected for Memory Forensics

DLL injection is one of the most common techniques used for code execution and persistence. Examining loaded modules provides valuable evidence of injected or hijacked libraries.

Unique Contribution

Sysmon records DLL loading events but does not maintain the current in-memory module state. Memory reveals which DLLs remain loaded after attack execution.

Forensic Intelligence Provided

- DLL injection
 - DLL hijacking
 - Reflective DLL loading
 - Suspicious unsigned modules
 - Process module inventory
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5. Windows.System.Handle

Description

The Handle artifact enumerates operating system object handles owned by running processes, including process handles, thread handles, token handles, file handles, registry handles, and synchronization objects.

Why Selected for Memory Forensics

Many attacker techniques require opening privileged handles before performing malicious operations. Handles therefore expose attacker intent and process relationships.

Unique Contribution

Sysmon records handle creation events only when specifically configured. Memory provides the actual handles that remain open during collection, enabling investigation of surviving process interactions.

Forensic Intelligence Provided

- LSASS access
 - Cross-process memory access
 - Token manipulation
 - Credential dumping indicators
 - Privileged object access
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6. Windows.System.Threads

Description

This artifact enumerates all execution threads belonging to active processes, including thread identifiers, start addresses, priorities, and execution information.

Why Selected for Memory Forensics

Many process injection techniques create malicious execution threads inside legitimate processes. Thread analysis is therefore essential for detecting injected execution.

Unique Contribution

Sysmon detects thread creation events, whereas memory identifies threads that remain active at collection time and their current execution state.

Forensic Intelligence Provided

- Remote thread injection
 - Suspicious thread start addresses
 - Memory execution anomalies
 - Process injection evidence
 - Thread ownership
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7. Windows.Memory.ProcessInfo

Description

ProcessInfo aggregates detailed information about running processes, including integrity levels, signer information, session identifiers, protection status, and execution attributes.

Why Selected for Memory Forensics

This artifact provides contextual information necessary for understanding the security posture and trust level of running processes.

Unique Contribution

Unlike basic process listings, ProcessInfo provides additional security metadata that assists forensic reasoning and evidence correlation.

Forensic Intelligence Provided

- Integrity levels
 - Protected processes
 - Signed vs unsigned executables
 - Privilege context
 - Security posture of processes
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8. Windows.Detection.Mutants

Description

The Mutants artifact enumerates Windows mutex objects created by running processes.

Why Selected for Memory Forensics

Many malware families create uniquely named mutexes to enforce single-instance execution, synchronize components, or indicate successful compromise.

Unique Contribution

Mutex objects rarely appear in traditional endpoint telemetry. Memory provides one of the few practical sources for observing active mutexes.

Forensic Intelligence Provided

- Malware fingerprints
 - Persistence indicators
 - Process synchronization
 - Family attribution clues
 - Active malware coordination
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9. Windows.Detection.Impersonation

Description

This artifact identifies processes executing under impersonated security tokens and extracts token-related privilege information.

Why Selected for Memory Forensics

Privilege escalation and token theft are common post-exploitation techniques. Token analysis directly reveals privilege abuse occurring within memory.

Unique Contribution

Sysmon records token-related events but cannot fully describe the current security token state of running processes. Memory exposes active impersonation contexts.

Forensic Intelligence Provided

- Token theft
 - Privilege escalation
 - SYSTEM impersonation
 - Stolen credentials
 - Security context analysis
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10. Windows.System.VAD

Description

The Virtual Address Descriptor (VAD) artifact enumerates the virtual memory layout of running processes, including memory protections, allocation types, backing files, and executable regions.

Why Selected for Memory Forensics

VAD analysis is one of the most powerful techniques in live memory forensics. It reveals executable memory regions, injected code, reflective DLLs, and unbacked executable pages that are invisible to traditional endpoint logs.

Unique Contribution

Unlike Sysmon, which records memory manipulation events, VAD captures the actual memory layout after those operations have occurred. It provides direct evidence of malicious memory modifications rather than only the triggering events.

Forensic Intelligence Provided

- RWX memory regions
- Private executable memory
- Unbacked executable regions
- Process injection evidence
- Reflective DLL injection
- Shellcode detection
- Memory execution anomalies